2026 Physics Prelim P1 Suggested Solutions

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	D	B	C	B	C	C	A	D	B	C	A	D	B	B	B
Q	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	B	D	C	D	B	A	C	A	A	C	B	B	B	C	A

1 D

Units of $\mu\frac{dT}{dt}$ = units of $I_0\cos(\omega t)$ = units of $\kappa(T-T_G)$

Units of μ = Unit of $\frac{I_0\cos(\omega t)}{(\frac{dT}{dt})} = \frac{(\text{J s}^{-1}\ \text{m}^{-2})(1)}{(\text{K s}^{-1})} = \text{J m}^{-2}\ \text{K}^{-1}$

$= (\text{kg m}^2\ \text{s}^{-2})(\text{m}^{-2}\ \text{K}^{-1}) = \text{kg s}^{-2}\ \text{K}^{-1}$

2 B

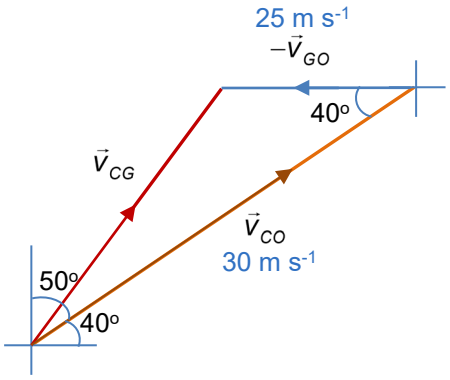
Let

- $\vec{v}_{CO}$: velocity of the cheetah
- $\vec{v}_{GO}$: velocity of the gazelle
- $\vec{v}_{CG}$: relative velocity of the cheetah wrt. to the gazelle

$\vec{v}_{CG} = \vec{v}_{CO} - \vec{v}_{GO}$

Hence, $v_{CG} = \sqrt{(25^2)+(30^2)-2(25)(30)\cos(40°)}$

$= 19\ \text{m s}^{-1}$



(Diagram not to scale)

3 C

Area under the acceleration-time graph is *change* in velocity, hence greatest velocity at 2T. Between 2T and 3T, while the acceleration is reversed and the car slows down, it is still moving farther away from its start point because the car has not reversed its velocity – at 3T it has the same velocity as at T.

4 B

<u>At the top of the trajectory</u>

The ball experiences an air resistance towards the left, opposite its velocity – it still has a horizontal velocity. Vertically, the experiences gravitational force acting downwards. Hence the resultant force is diagonally down and to the left.

Hence acceleration is diagonally down and to the left.

5 C

The platform-man system is in equilibrium, i.e. resultant force = zero.

Weight of platform-man = Force by water on platform-man

By Newton's 3rd Law, |Force by water on platform-man| = |Force by platform-man on water|

By Newton's 2nd Law, Force by platform-man on water = $\frac{\Delta p}{\Delta t} = \frac{\Delta(mv)}{\Delta t} = \frac{\Delta m}{\Delta t} \times \text{v} = 40v$ (where v is the velocity of water leaving the platform).

$mg = 40v$

$v = \frac{(96)(9.81)}{40} = 23.54 = 24\ \text{m s}^{-1}$ (2 s.f)

6 C Impulse = change in momentum = final momentum – initial momentum
Taking the direction away from the player as the positive direction,
$+6.7 = 0.45v_f - 0.45(-3.2)$
$v_f = 12$ m s^{-1}

7 A Since the system is in equilibrium, the resultant force *R* must have the same magnitude as the weight and its direction must be vertically upwards. Note that T_2 is larger than T_1; the length of the string is not an indication of the magnitude of the force.

8 D By the principle of conservation of energy,

Loss in GPE = Work Done by (mean) force on spike (as it enters the ground)

mg(h+x) = *Fx* (where *m* is the mass of spike, *h* is 4.5 m, *x* is 0.25 m)

$$F = \frac{mg(h+x)}{x} = \frac{(3.0)(9.81)(4.5+0.25)}{0.25} = 560 \text{ N (2 s.f)}$$

9 B When the motor is not spinning, the 60 N mass will move downwards as there is a net downward force of 35 N.
Since the belt remains stationary when the motor is spinning,
$P_{output} = F_{net} \times$ (distance moved per unit time) $= F_{net} \times (20\pi d)$
$= 35 \times (20 \times \pi \times 0.35) = 770$ W

10 C Tension *T* provides centripetal force.
$$T = \frac{mv^2}{r} = \frac{(0.30)(4.0)^2}{0.60} = 8.0 \text{ N}$$
Net force *F*, due to weight *W* and tension *T*

$$F = \sqrt{T^2 + W^2} = \sqrt{8.0^2 + (0.30 \times 9.81)^2} = 8.52 \text{ N}$$

11 A At point P,
Gravitational field strength due to M = $\frac{GM}{x^2}$
Gravitational field strength due to 9*M* = $\frac{G9M}{y^2}$
As resultant gravitational field strength = 0,

$$\frac{GM}{x^2} = \frac{G9M}{y^2}$$
$$\frac{1}{x^2} = \frac{9}{y^2} \Rightarrow \text{x : y = 1 : 3}$$

12 D At point Q where the gravitational field strength g is zero, since $g = -\frac{d\phi}{dr}$, the gravitational potential gradient must be zero.

Options A and B are wrong because the gravitational potential (and hence GPE of a mass placed there) is the largest (least negative) at Q.

Option C is wrong because the gravitational potential at a point is independent of the mass that is at that point; it is just due to the two planets.

13 B Gradient of *a*-*x* graph is $-\omega^2$.

P's ω^2 is four times that of Q's. P's amplitude is half of Q's.

Given $E_K = \frac{1}{2}mv_0^2 = \frac{1}{2}m\omega^2 x_0^2$, the maximum kinetic energy (this is also called the energy of oscillation) of P is the same as that of Q's.

14 B Increase in EPE = $\frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2 = \frac{1}{2}(25)\left(0.070^2 - 0.055^2\right) = 0.0234$ J

(The increase in EPE is NOT $\frac{1}{2}k\left(x_f - x_i\right)^2$.)

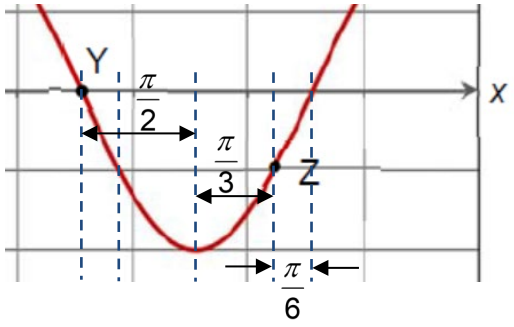
Energy of oscillation = work done by external force to pull it down
= increase in EPE – loss in GPE
= 0.0234 - $(0.140)(9.81)(0.015)$
= 0.00280 J

Note that this energy of oscillation = maximum kinetic energy of the mass. It can also be calculated from $\frac{1}{2}m\omega^2 x_0^2 = \frac{1}{2}kx_0^2$, since $\omega^2 = \frac{k}{m}$.

$\frac{1}{2}kx_0^2 = \frac{1}{2}(25)(0.015)^2 = 0.00281$ J

15 B Air particle at point X is at a point of compression and hence its displacement is zero.

Option A is wrong because nodes and antinodes apply only to stationary waves.
Option C is wrong because in a progressive wave every particle has the same amplitude.
Option D is wrong. The phase difference is $\frac{5\pi}{6}$.



16 B When the sliding section of Path I is physically pushed inwards by a distance of 2.5 cm, the path length of Path I is changed by 5.0 cm = 0.050 m.

DI to CI is half a wavelength → $\frac{\lambda}{2} = 0.050$ m $\Rightarrow \lambda = 0.10$ m

Hence, frequency of the wave, $f = \frac{v}{\lambda} = \frac{340}{0.10} = 3.4$ kHz

17 D For a standing wave on a string fixed at both ends, the resonance frequencies are given by : $f_n = \frac{nv}{2L}$

For the 50 cm string, if resonance occurs : $f_n = \frac{n220}{2(0.50)} = 440 \Rightarrow n = 2$

The 50 cm string can cause the tuning fork to resonate.

For the 40 cm string: $f_n = \frac{n220}{2(0.40)} = 275n = 440 \Rightarrow n = 1.6$.

Hence, 440 Hz is not one of the harmonics produced by the 40 cm string, hence it fails to resonate the tuning fork.

(Quick check for 40 cm string)

- Fundamental frequency (n=1): 275 Hz ;
- Second harmonic (n = 2): 275 x 2 = 550 Hz.

18 C $Q = mc\Delta\theta$

$P = mc\frac{\Delta\theta}{t}$

$P = mc(\frac{4}{60})$

$Q = mL$

$P(40 \times 60) = mL$

Hence,

$c(\frac{4}{60}) = \frac{L}{40 \times 60}$

$\frac{c}{L} = \frac{1}{160} = 0.0063 \text{ K}^{-1}$

19 D $p = \frac{1}{3}\frac{Nm}{V} < c^2 >$ [this is in formula list]

$1.00 \times 10^5 = \frac{1}{3}(1.60) < c^2 >$

$< c^2 >= 187500$

Root-mean-square speed = 433 m s^{-1}

20 B $\frac{kq}{2.0} + \frac{k(2q)}{2.0} = \frac{3kq}{2.0} = 6740$

The electric field at the mid-point due to the positive and negative charge are in the same direction.

$\frac{kq}{2.0^2} + \frac{k(2q)}{2.0^2} = \frac{3kq}{4.0}$

$= \frac{6740}{2}$

$= 3370 \text{ N C}^{-1}$

21 A

Equating $I = nqv_d A$ and $I = \frac{V}{R} = \frac{V}{\rho L} A$,

$$v_d = \frac{V}{\rho L n q} \propto \frac{1}{L}$$

We need to find the new length of the conducting putty.

The volume is conserved, so $AL = (\frac{A}{4})L' \rightarrow L' = 4L$

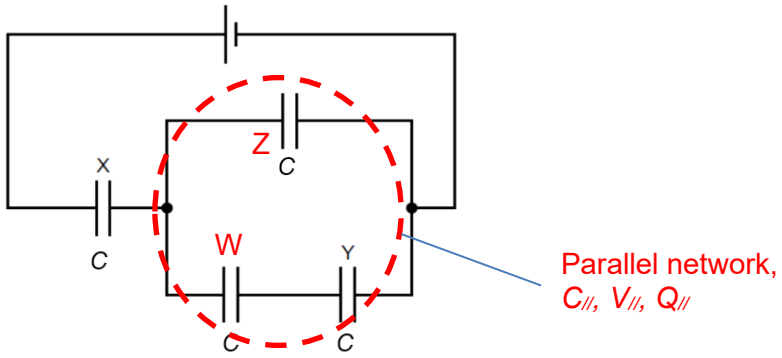
Hence $\frac{v_d'}{v_d} = \frac{L}{L'} = \frac{1}{4}$

22 C Total energy that can be supplied by the battery E = (12 V)(60 Ah)

Total energy that is used by the lamp = (36 W)(t)

Hence, the maximum time t = (12)(60)/(36) = 20 h

23 A



For ease of reference, let's label all the capacitors as above.

The equivalent capacitance of the parallel network of capacitors on the right:

$$C_{//} = C + (\frac{1}{C} + \frac{1}{C})^{-1} = \frac{3}{2}C$$

Capacitor X is in series with the whole right-hand network, so $Q_{//} = Q_X$ = CV.

Hence, $V_{//} = \frac{Q_{//}}{(\frac{3}{2}C)} = \frac{Q_X}{(\frac{3}{2}C)} = \frac{CV}{(\frac{3}{2}C)} = \frac{2}{3}V$

In the lower branch of the parallel circuit, C and C are in series which means they carry the same charge. Same charge and same capacitance → same potential difference.

$V_Y = = \frac{V}{3}$

24 A When the switch is closed, the capacitor charges up until it reaches the e.m.f. of the battery V_0.

P.d. across the capacitor increases with time: $V_c = V_0(1 - e^{-\frac{t}{RC}})$

Hence, the p.d. across the resistor decreases with time : $V_R = V_0 e^{-\frac{t}{RC}}$

[the p.d. across the resistor was initially the e.m.f. of the battery, and the current in the circuit is V_0/R.]

Power dissipated by the resistor, $P = \frac{V_R^{\,2}}{R} = \frac{V_0^{\,2}}{R} e^{-\frac{2t}{RC}}$

Or simply consider that the current varies in the circuit (and therefore through the resistor) as: $I_R = I_0 e^{-\frac{t}{RC}}$ and the power in R is $P = I_R^{\,2} R = I_0^{\,2} R e^{-\frac{2t}{RC}}$

The power in *R* follows an exponential decay, hence there is a constant "half-life" (just as in radioactive decay).

Option A shows an **exponential decay of power with a constant half-life**.

25 C The magnetic force acting on the proton is upwards. To balance this force, so that the proton would pass undeflected, the electric force on the proton acts downwards, so the electric field is directed downwards.

$Bqv = qE$

$v = \frac{E}{B} \rightarrow E = Bv = (0.015)(2.0 \times 10^7) = 3.0 \times 10^5 \text{ N C}^{-1}$

Note that the sign and magnitude of the charge do not matter; if the speed is right, the particle is undeflected. If it were a negative charge the electric force and magnetic force directions just swops.

26 B By the principle of conservation of energy,
Gain in KE of particle = Loss in Electric Potential Energy
$\frac{1}{2} mv^2 - 0 = qV$

$$v = \sqrt{\frac{2qV}{m}} \text{ ---------- (1)}$$

As magnetic force provides centripetal force,

$$Bqv = \frac{mv^2}{r}$$

$$r = \frac{mv}{Bq} \text{ ---------- (2)}$$

Substitute (1) into (2):

$$r = \frac{m}{Bq}\sqrt{\frac{2qV}{m}} = \frac{1}{B}\sqrt{\frac{2mV}{q}}$$

Since B and V are the same for both,

$$r \propto \sqrt{\frac{m}{q}}$$

$$\frac{r_Y}{r_X} = \sqrt{\frac{m_Y q_X}{m_X q_Y}} = \sqrt{\frac{m_Y}{m_X} \times \frac{q_X}{q_Y}} = \sqrt{2 \times \frac{1}{4}}$$

Therefore, $r_Y = \frac{r_X}{\sqrt{2}}$

27 B *F = Magnetic force on YZ + Magnetic force on XY*

$$F = F_{XY} + F_{YZ}$$

As the component of L_{XY} which is perpendicular to the magnetic field is the same as L,

$$F_{XY} = F_{YZ} = \frac{F}{2}$$

$$F_{XY} = \frac{F}{2} = BIL_{XY}\cos\theta \text{ (as } L = L_{XY}\cos\theta\text{)}$$

$$L_{XY} = \frac{F}{2BI\cos\theta}$$

28 B Electron can excite gas to n = 4 only, since $E_4 - E_1 = -1.52 - (-5.13) = 3.61 \text{ eV}$ but $E_5 - E_1 = -1.38 - (-5.13) = 3.75 \text{ eV}$.

De-excitation	Energy of emitted photon	Larger than work function?
4 → 1	3.61 eV	Yes
4 → 2	1.51 eV	Yes
4 → 3	0.42 eV	No
3 → 1	3.19 eV	Yes
3 → 2	1.09 eV	No
2 → 1	2.10 eV	Yes

29 C The type of emission is beta since the count rate drops significantly after aluminium is placed between the source and GM tube.

Based on the decrease in count rate, 3 half-lives have passed. Hence half-life is 8 hours.

30 A For an alpha decay, nucleon number decreases by 4, proton number decreases by 2.
For a beta decay, nucleon number remains unchanged, proton number increases by 1.

Thus for 3 alpha decays and 1 beta decay, nucleon number will decrease by 12 and proton number will decrease by 5.